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|---|---------|---|----------|-----|
| 2 | (a) (i) | point at which whole weight of body
may be considered to act | M1
A1 | [2] |
| | (ii) | sum of forces in any direction is zero
sum of moments about any point is zero | B1
B1 | [2] |
| | (b) | <i>either:</i>
T and W have zero moment about P
so F must have zero moment, i.e. pass through P
<i>or:</i>
if all pass through P, distance from P is zero for all forces
so sum of moments about P is zero | M1
A1 | [2] |
| | (c) (i) | $F \cos \alpha = T \cos \beta$ | B1 | [1] |
| | (ii) | $W = F \sin \alpha + T \sin \beta$ | B1 | [1] |
| | (iii) | $2W = 3T \sin \beta$ | B1 | [1] |
| 3 | (a) | sum of (random) kinetic and potential energies | M1 | |